

Sports Platform - Business & Technical Documentation

Problem Statement

Millions of sports enthusiasts struggle to find and access sports infrastructure, players, coaches, and tournaments through a unified platform. Existing solutions focus on individual aspects such as bookings or fitness tracking, resulting in fragmented experiences. Ground owners face low utilization rates, tournament organizers struggle with participant acquisition, and players lack personalized insights into their sports performance. The absence of a centralized ecosystem limits sports participation, community engagement, and grassroots talent development.

Solution

Our solution is an AI-powered sports ecosystem that combines infrastructure discovery, venue booking, tournament participation, player networking, and fitness analytics into a single platform. Users can discover grounds, connect with players, join tournaments, and track sports-specific performance. Integration with smartwatches enables advanced fitness tracking, while AI generates recommendations, insights, and personalized improvement plans. The platform also provides value to venue owners and organizers through booking management and analytics dashboards.

Target Audience

Casual players, fitness enthusiasts, amateur athletes, professional athletes, coaches, sports academies, tournament organizers, and sports facility owners.

Microservices Architecture

API Gateway, Auth Service, Profile Service, Venue Service, Booking Service, Match Service, Tournament Service, Player Service, Wearable Service, Analytics Service, Recommendation Service, Notification Service, Payment Service, RabbitMQ Event Bus, PostgreSQL, Redis.

Technology Stack

Frontend: React Native (Android, iOS, Web), TypeScript, Expo. Backend: FastAPI, Python. Database: PostgreSQL. Cache: Redis. Messaging: RabbitMQ. Authentication: JWT. Hosting: Render. Storage: Cloudinary. Analytics & AI: Python, Pandas, Scikit-Learn.

Business Model

Venue booking commissions, tournament registration fees, premium athlete subscriptions, advertising partnerships, academy listings, venue management subscriptions, and B2B analytics services.

Future Scope

AI coaching, injury prediction, talent discovery, sports scouting, sponsorship matching, community-driven leagues, wearable intelligence, and predictive analytics for sports infrastructure utilization.